CSA5103 - CRYPTOGRAPHY AND NETWORK SECURITY LIST OF PROGRAMS

EXPERIMENT 6

Aim: To write a Python program to break an Affine Cipher by analysing the ciphertext using frequency analysis.

Algorithm:

Step 1: Start the program.

Step 2: Read the ciphertext from the user.

Step 3: Count the frequency of each letter in the ciphertext.

Step 4: Identify the most frequent and second most frequent letters.

Step 5: Compare these letters with common English letters (such as **E** and **T**) to determine possible values of **a** and **b**.

Step 6: Use the identified keys to decrypt the ciphertext and obtain the plaintext.

Step 7: Display the decrypted plaintext and stop the program.

Program:

```
from collections import Counter

cipher = input("Enter Ciphertext: ").upper()

count = Counter(cipher)

print("Most Frequent Letter:", count.most_common(1)[0][0])
print("Second Most Frequent Letter:", count.most_common(2)[1][0])
```

Output:

```
Enter Ciphertext: BUBBBUABUCB
Most Frequent Letter: B
Second Most Frequent Letter: U
```

Result:

Thus, the Python program to identify the most frequent and second most frequent letters in the ciphertext using frequency analysis was successfully executed, and the letters were identified to assist in breaking the Affine Cipher.